

# ACI 544-C: Notched vs. Unnotched Methods of Testing Fiber-Reinforced Concrete (FRC) in Flexure

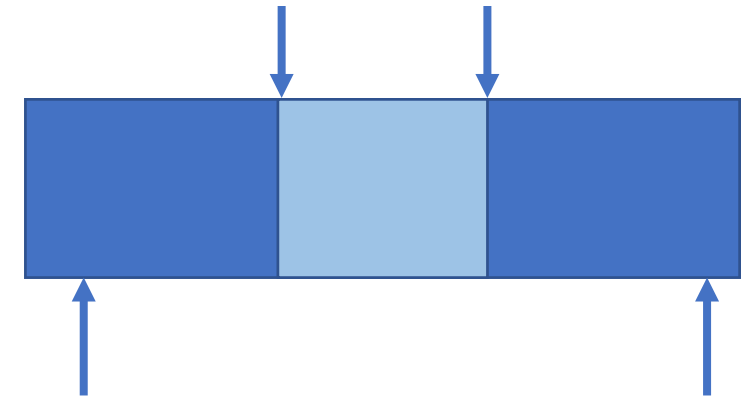
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**Arizona State University, Tempe, Arizona**

**March 30<sup>th</sup>, 2021**  
**ACI Convention**

# Introducing a Notch Test option to the ASTM C1609 test and its effects

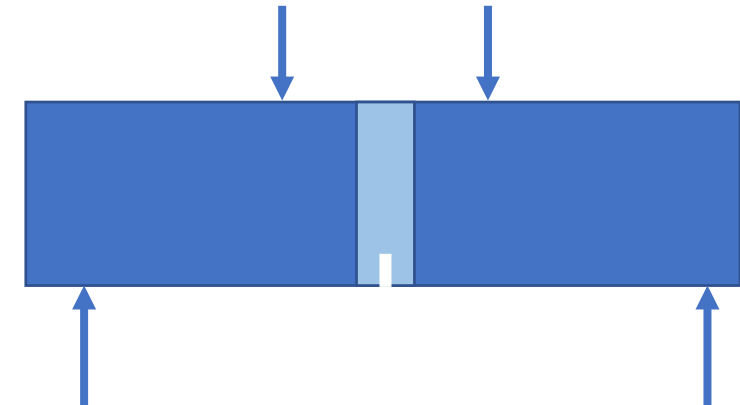
## Objectives:

- ASTM Does not have a procedure to test a notched beam
- RILEM TC 162-TDF : Test and Design Methods For Steel Fibre Reinforced Concrete
- EN-14651
- To address difference in both tests, failure mechanisms, material properties and appropriate test for design procedures.

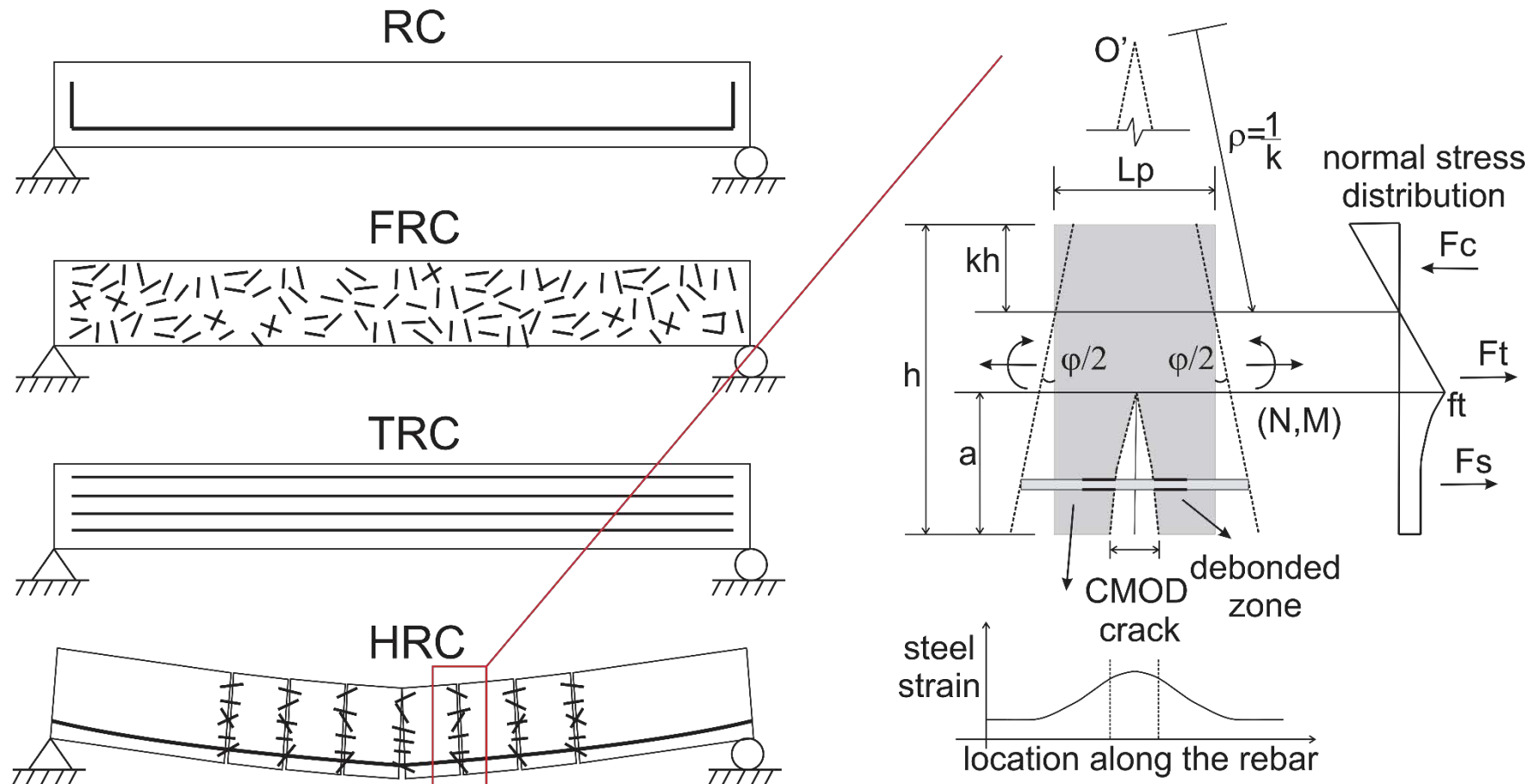


Notch reduces the first peak and variability in test data.

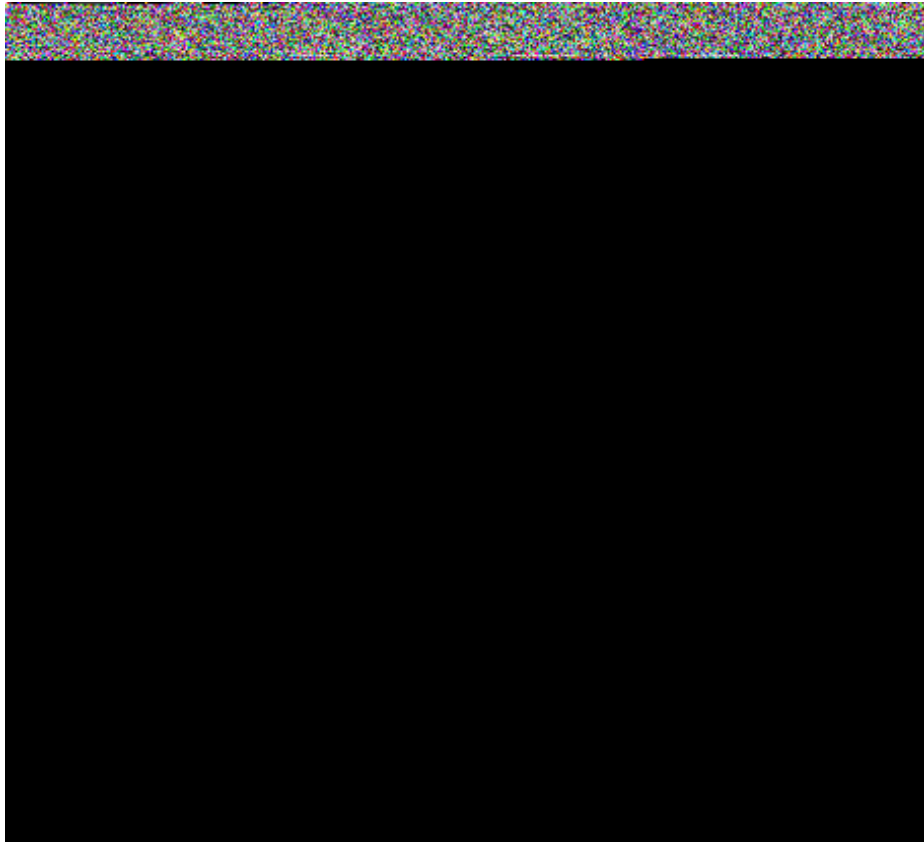
- Effect on the first peak
- Effect on the multiple cracking and localization effects
- Effect on the back calculated material parameters and the residual strength parameters
- Appropriate test to evaluate design stress-strain parameters



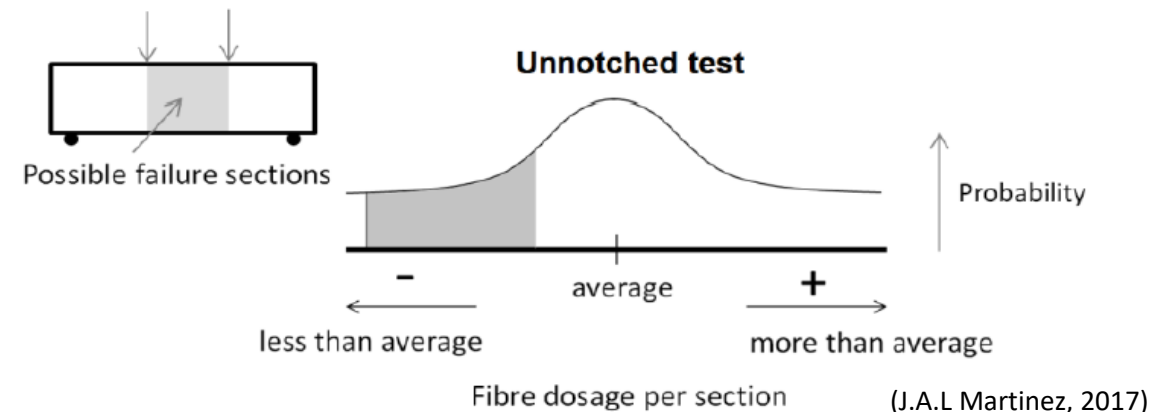
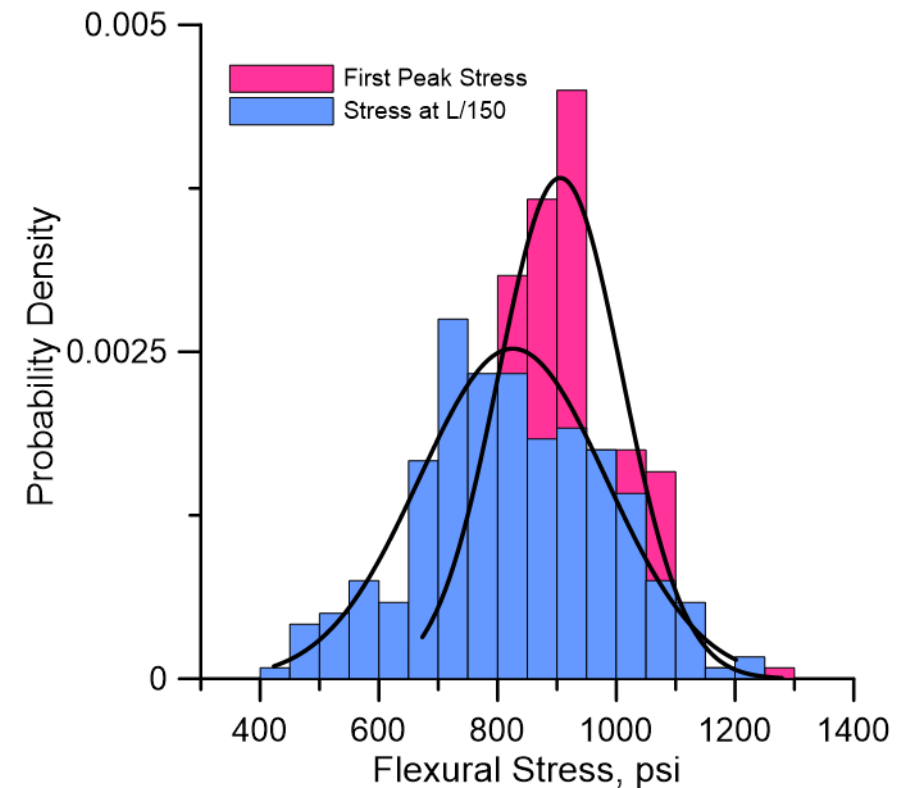
# ASU Localized hinge Model



## ASTM C1609 Data: What we've learned so far

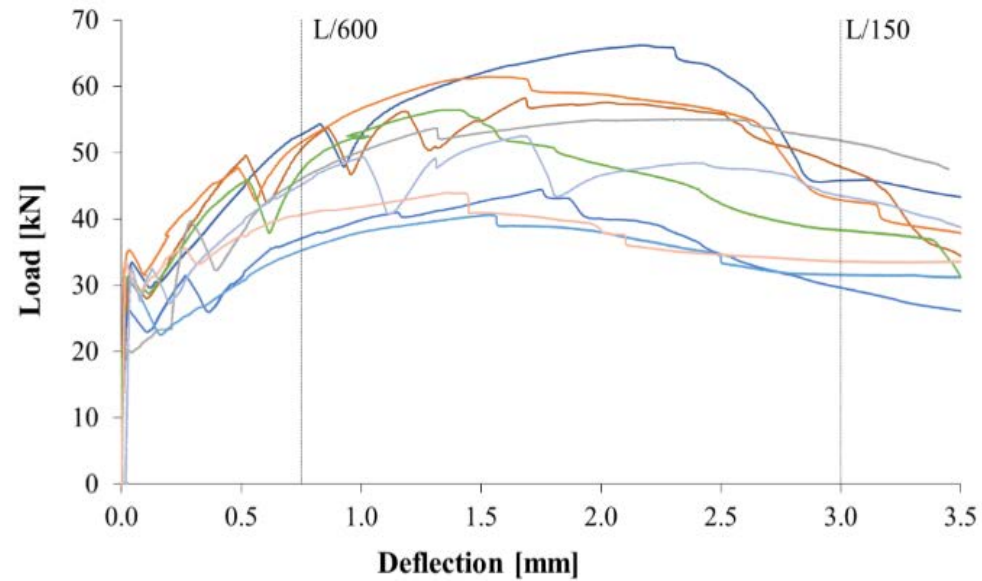


- Lower Variability in MOR compared to other parameters as crack will form at the weakest link inside the constant moment zone.
- Higher stress at MOR
- Higher MOR can reduce overall response of the fiber

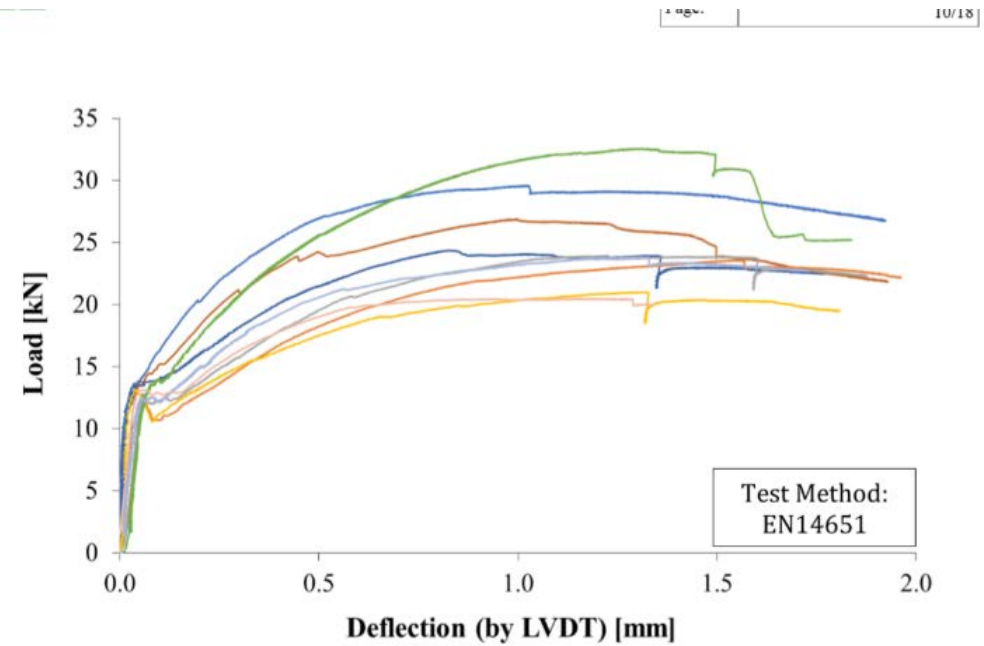


weakest link will have less fiber dosage than average

# SFRC- Comparison of notched and unnotched beams, ASTM C1609 vs. EN14651

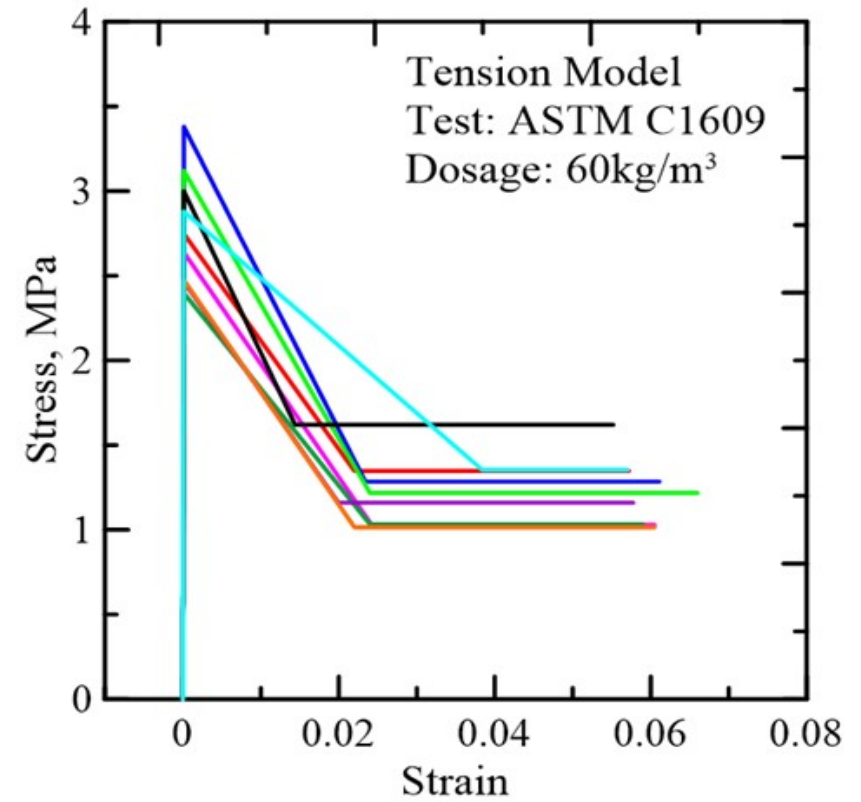
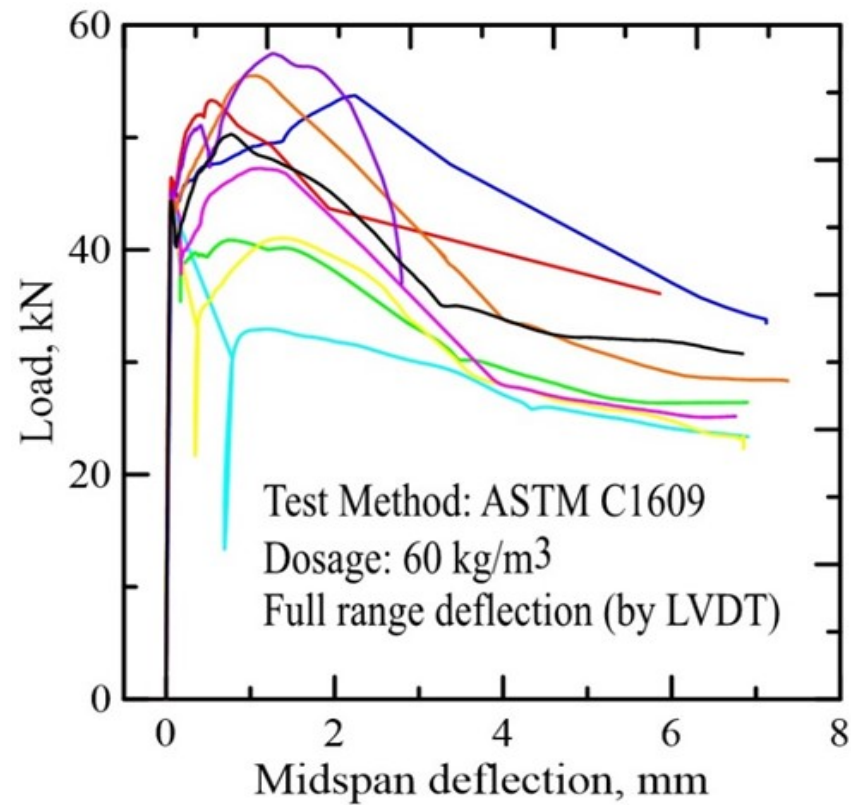


ASTM C1609



EN14651

Multiple Samples provide the range of properties

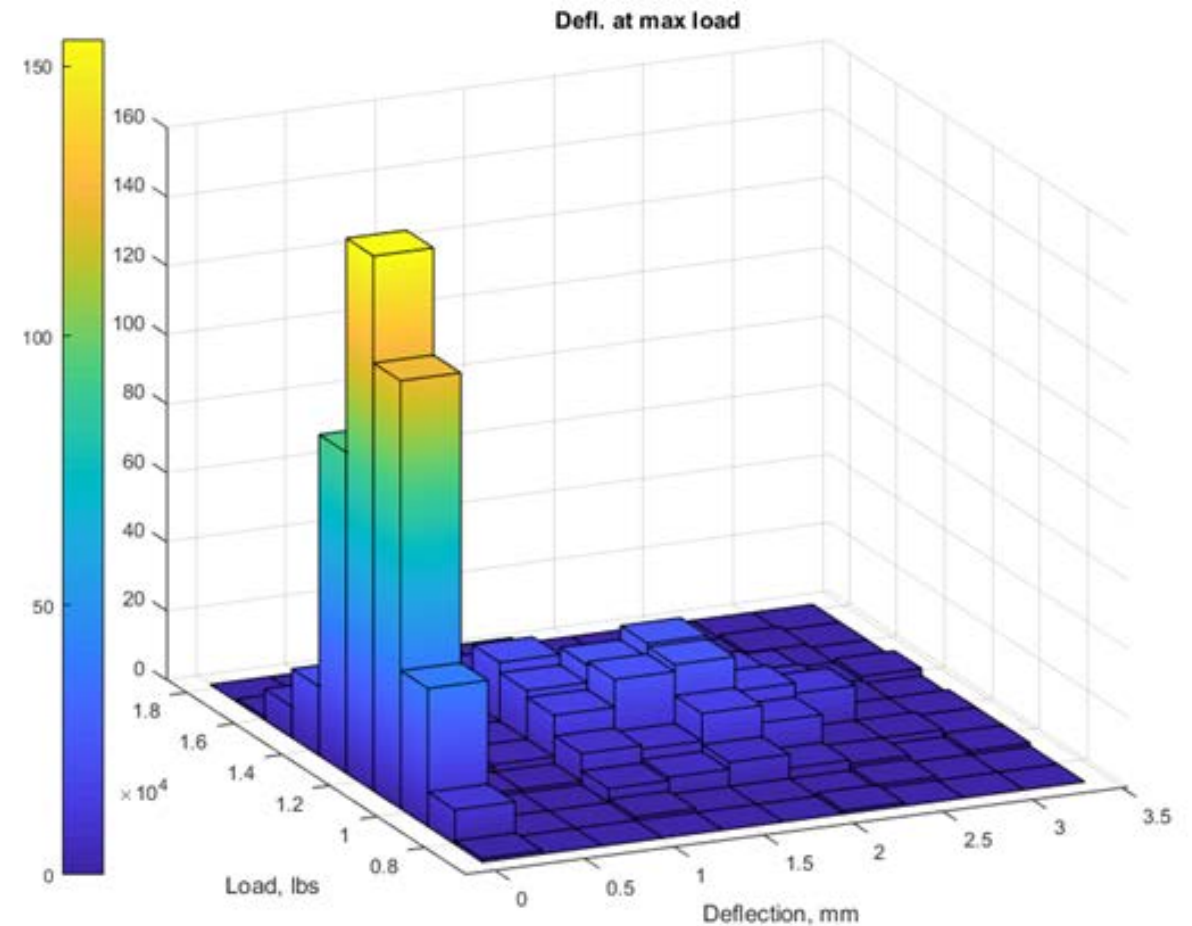


# MOR dominates the FRC response

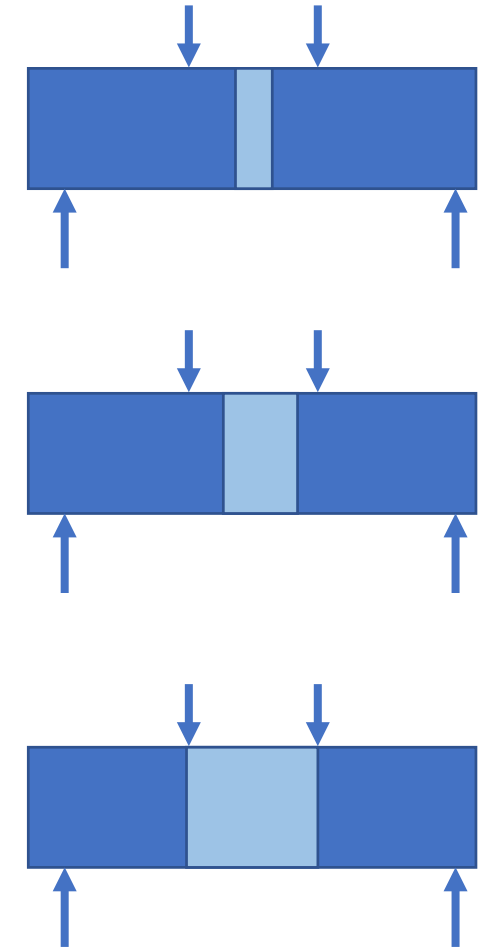
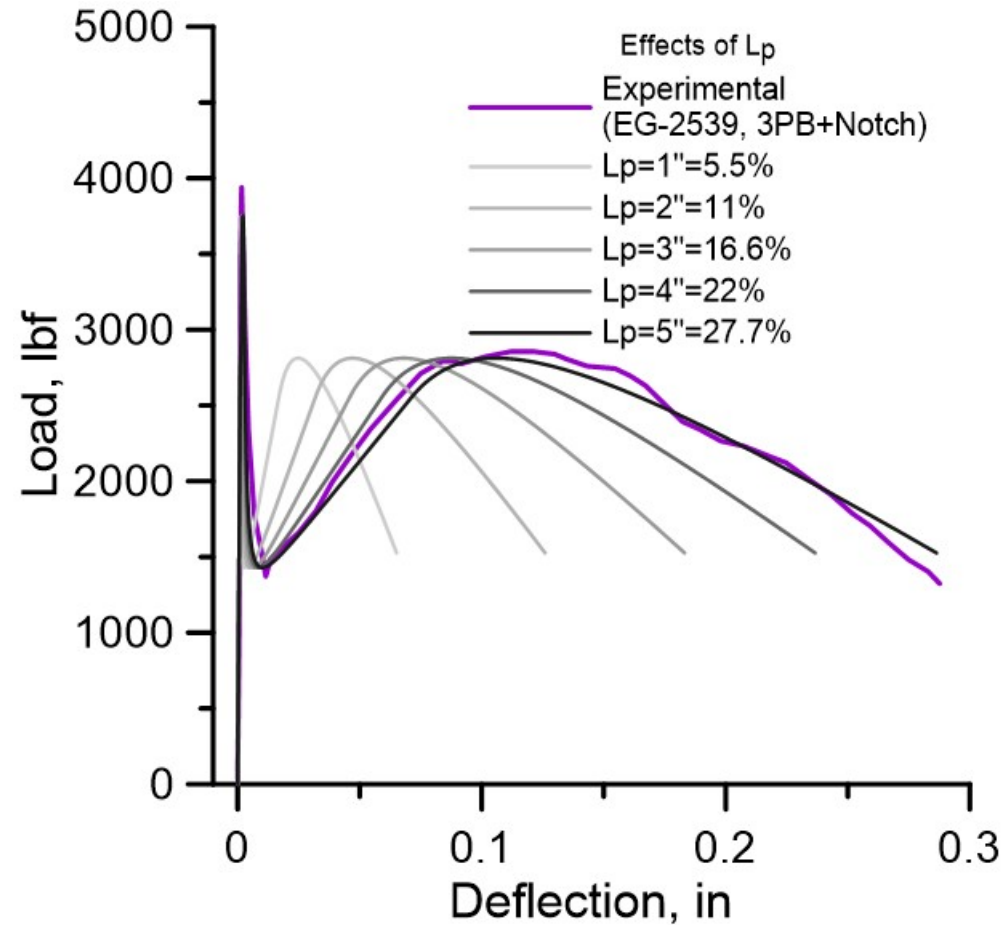
800+ C1609 test data show that max load for most test was the MOR (0.5%  $V_f$  SFRC)

Do not use it for normalization though

For deflection hardening, confusion ensues



# Localization Zone effect in the post peak response

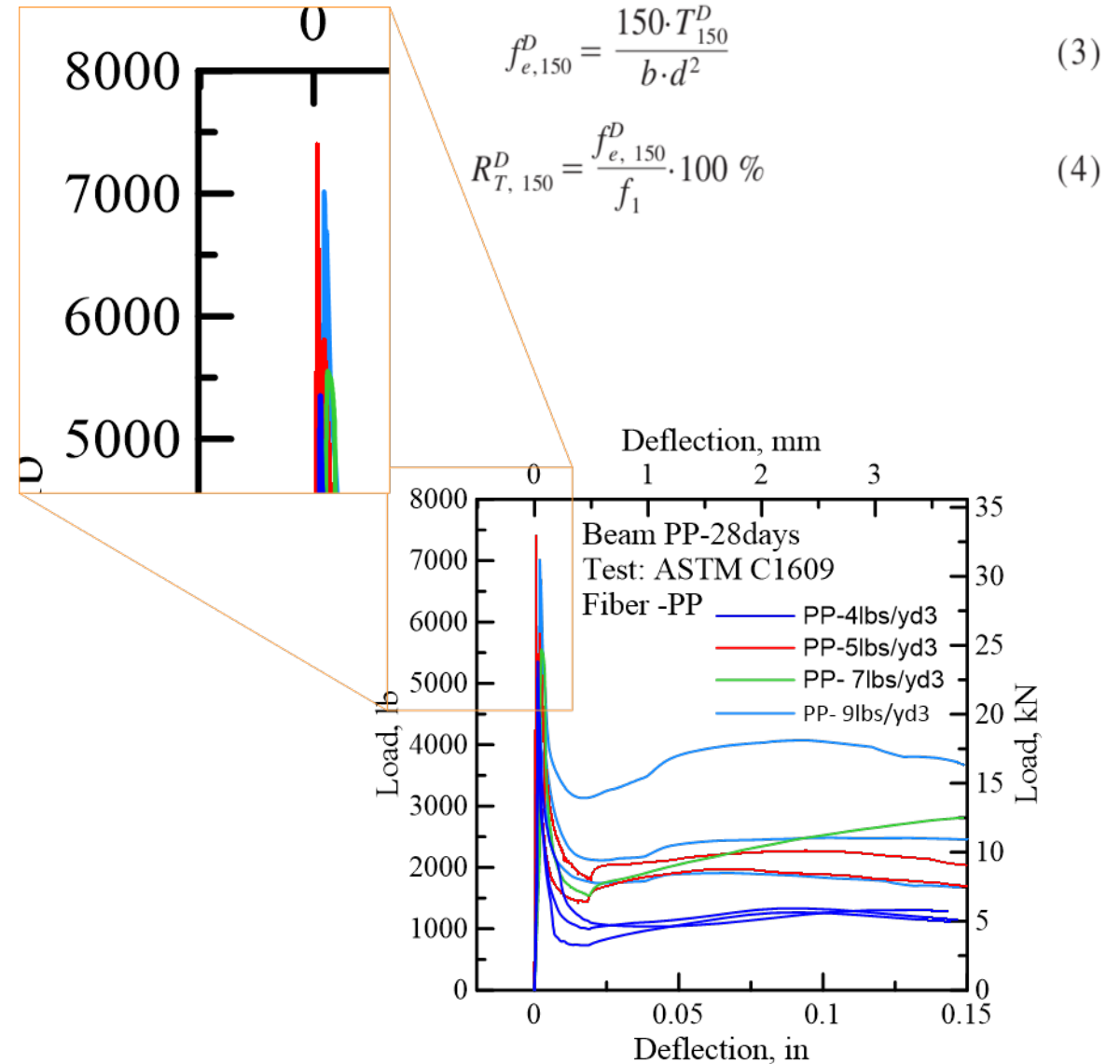


# Variability in reporting the residual strength and MOR

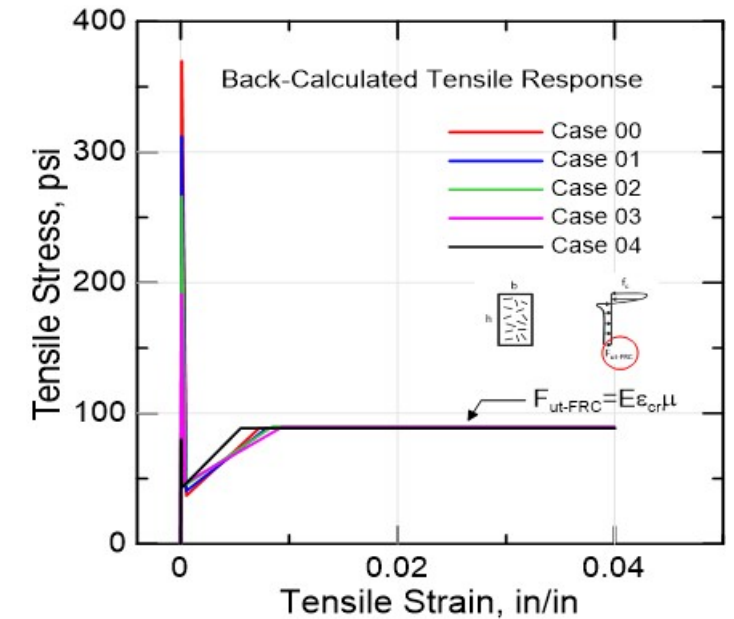
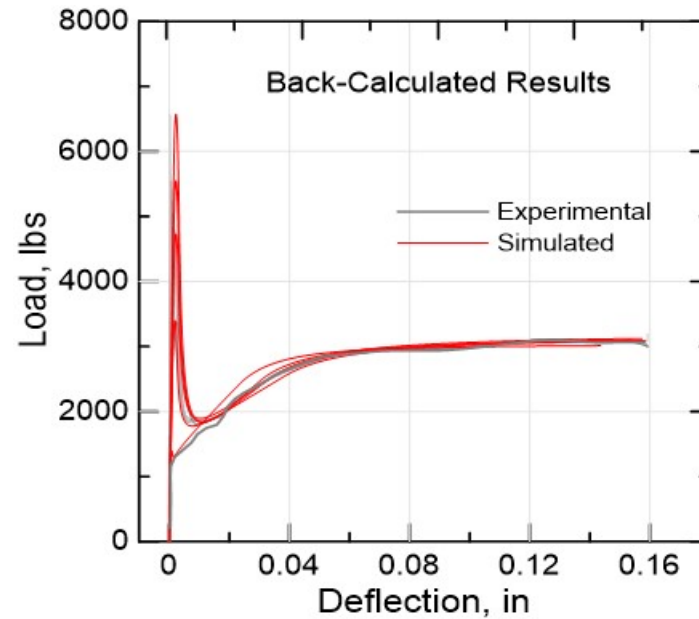
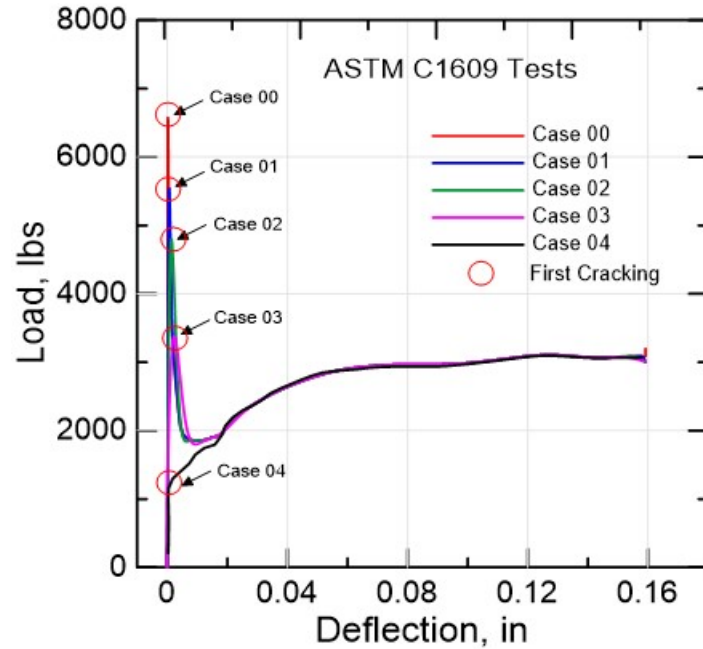
$R_{T,150}$  normalizes with first peak stress which makes it more variable and is not significant to the actual residual strength.

MOR will reveal less about the ductility of the mixture and its residual strength, and more about the possibility of a random flaw in an unstable test protocol

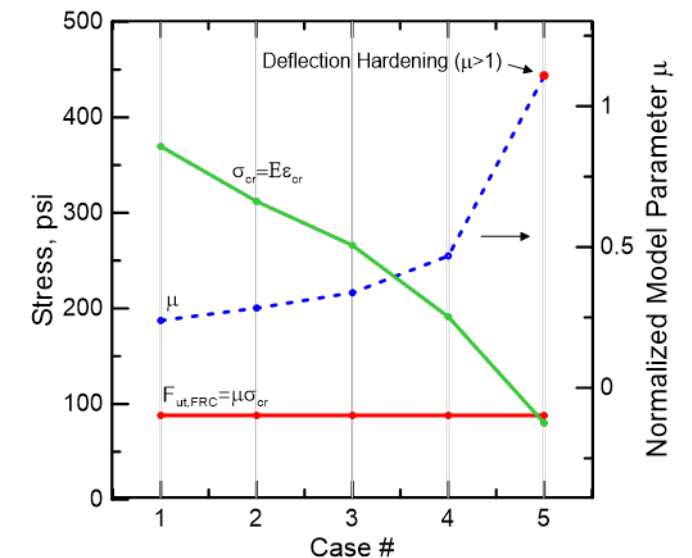
Our Normalization parameter will make all the test results unstable, unreliable



## Effect of MOR on ACI Design Approach, ACI 544-8R

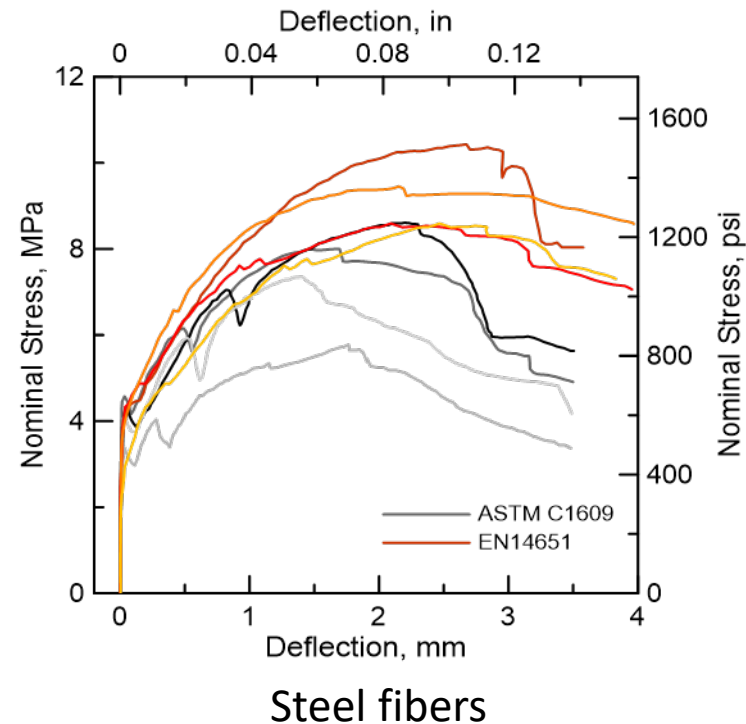
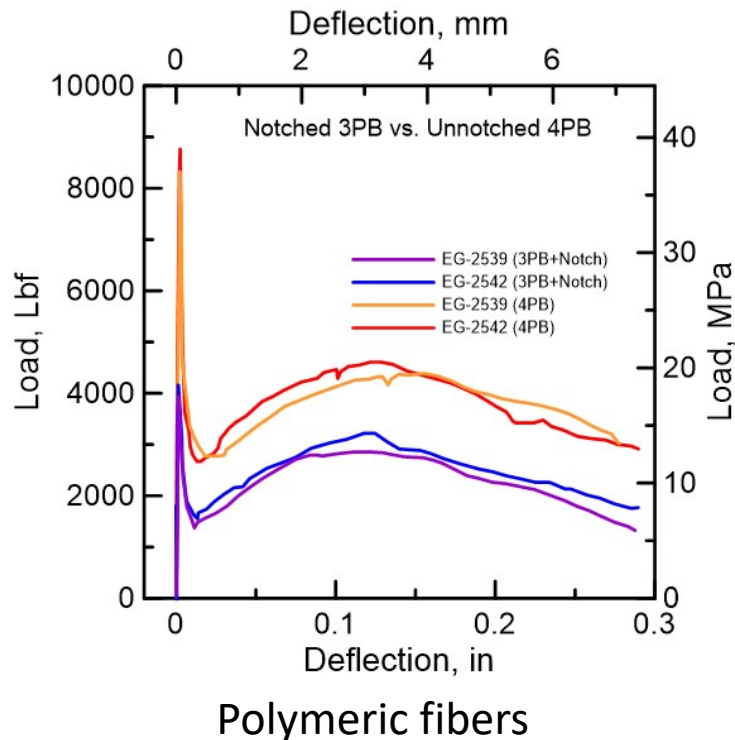
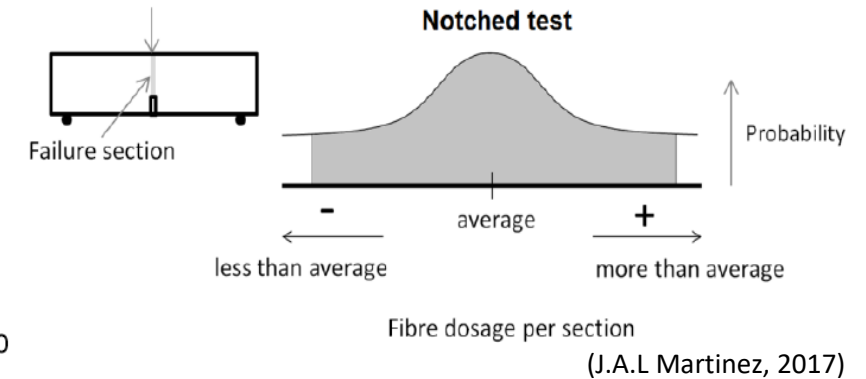
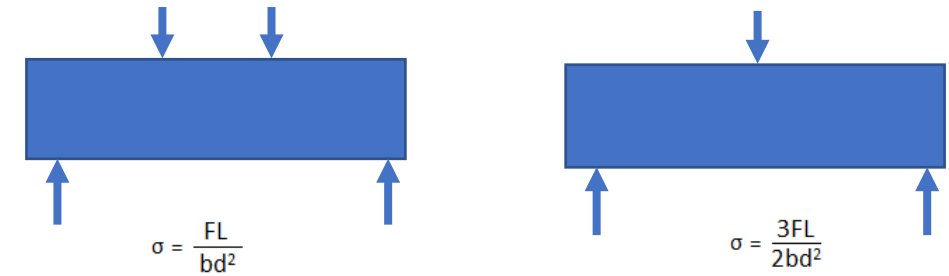


- Multi-point smoothing and averaging of the data will affect the peak load and the high MOR values drop significantly.
- Even though parameters are normalized with cracking point, back-calculated residual tensile strength by ACI-544-8R is independent of the varying peak.
- Use the residual load as a measure of post crack response and normalize it appropriately, reporting as equivalent flexural strength is misleading.



## 4PB vs. 3PB notched

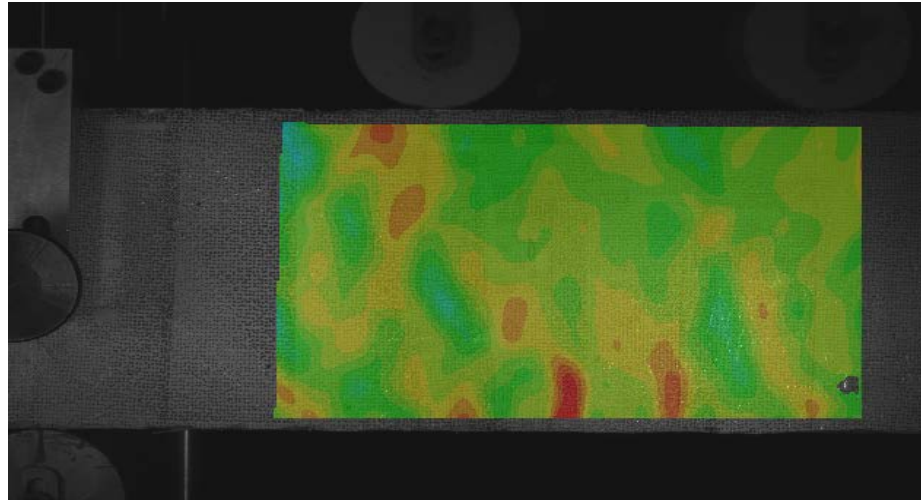
- For notched, MOR will be less due to stress concentration, but more variable compared to unnotched as the zone is restricted to a notch which can be either weak or strong segment.
- Maximum deflection won't be at the center leading to lower deflection measurements



Non-central cracking in unnotched

## DIC Data showing strain distribution before localization

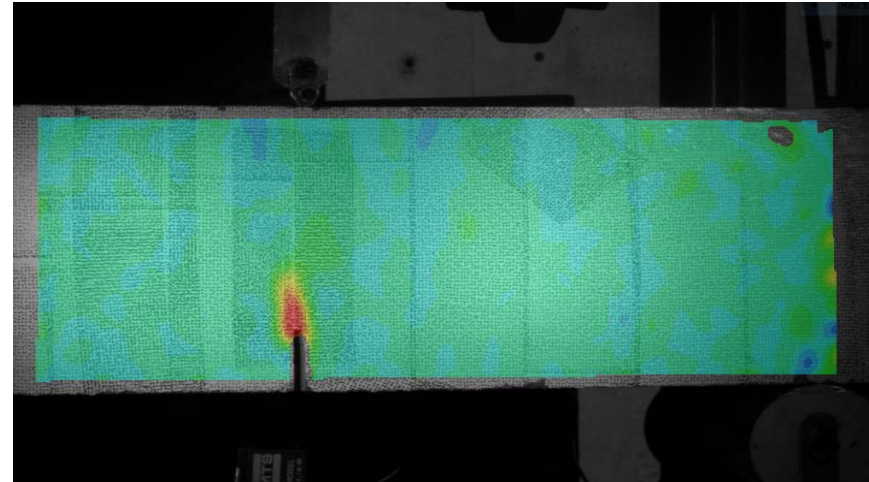
Unnotched



Multiple crack initiation

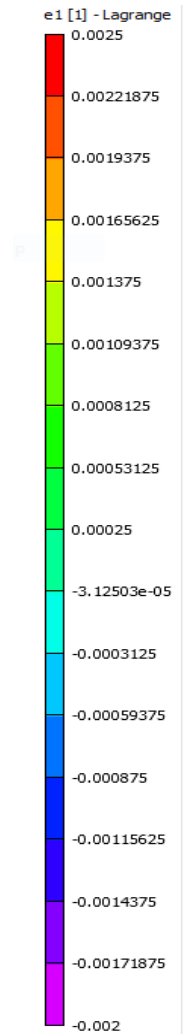
- Multiple Cracking
- Non-central localization
- Weakest link

Notched



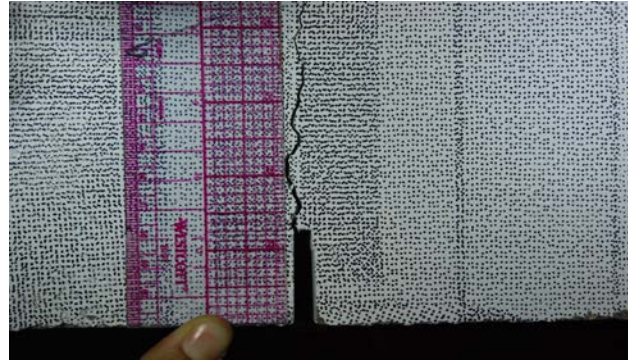
Single localized cracking

- Single Crack properties
- Central cracking
- Not the weakest link

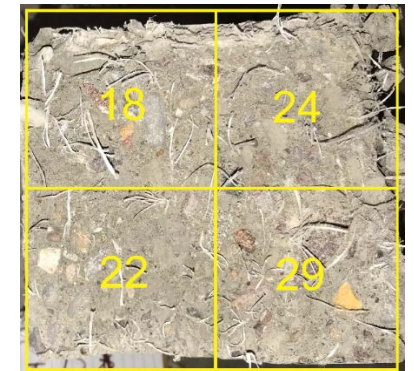


# Difference in crack formation

## Notched

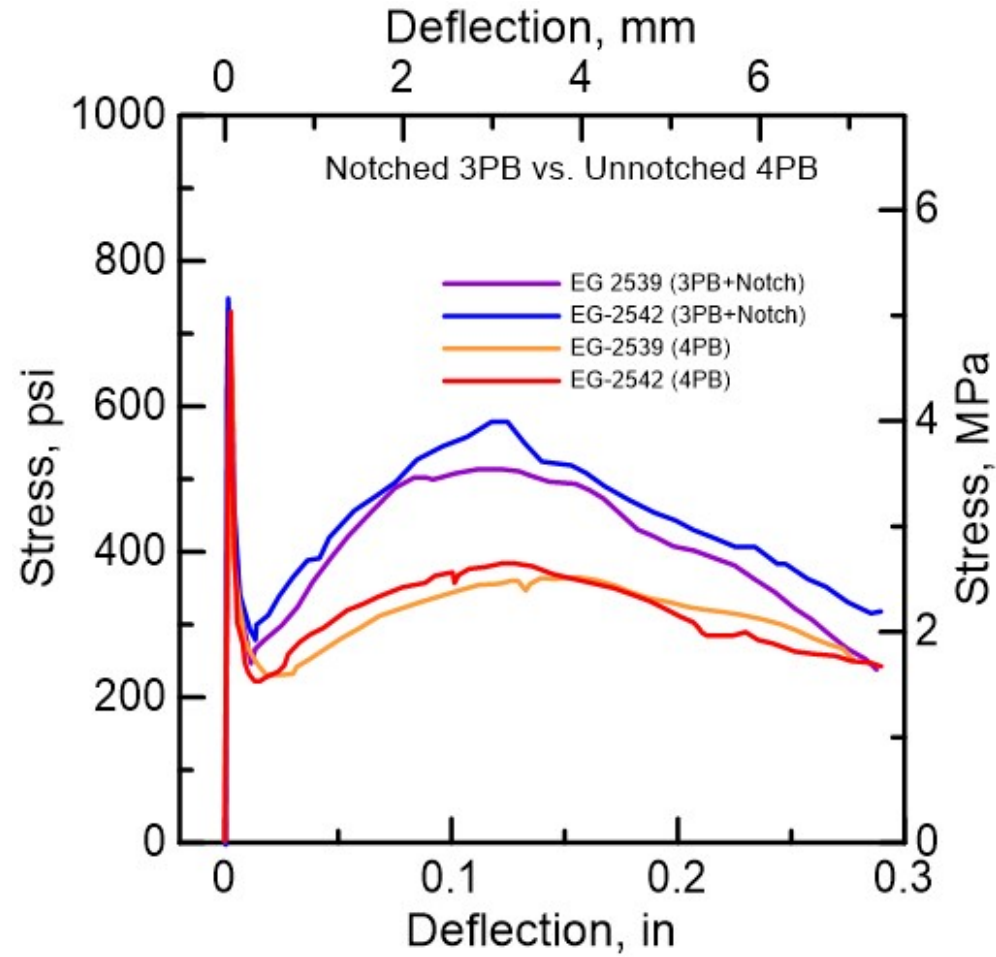


## Unnotched



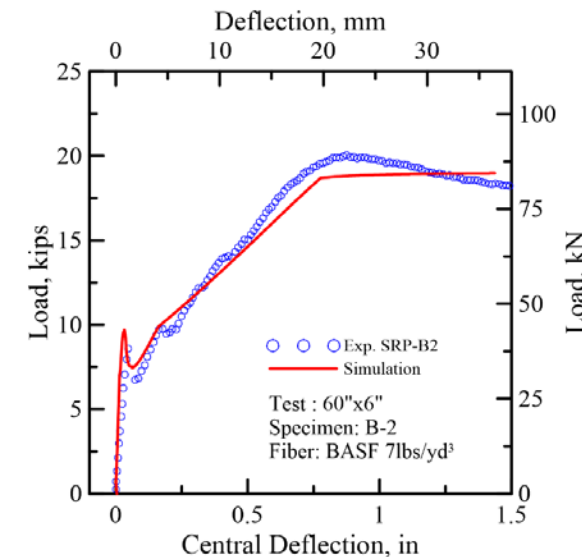
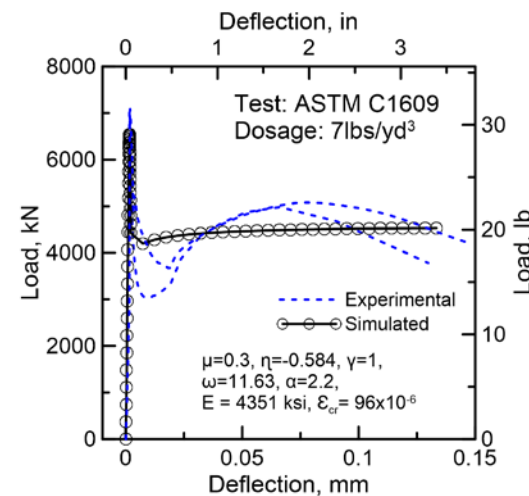
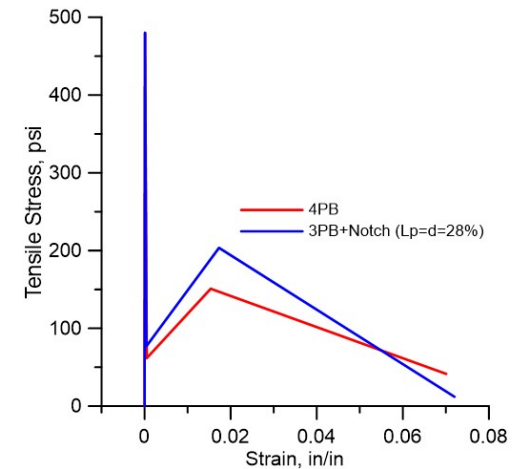
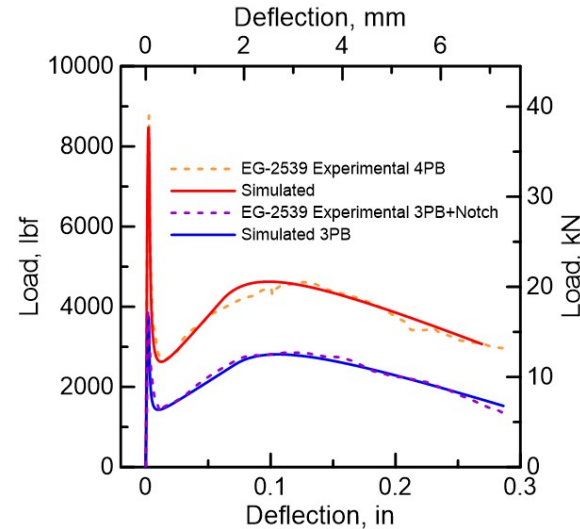
When characterizing FRC, test data should include pictures, crack location and number of fibers

# Stress calculations



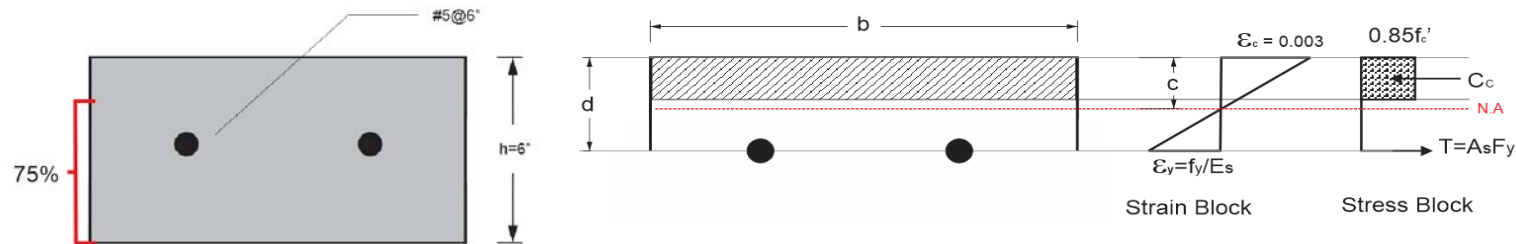
# Back-calculation Procedure for Tensile Properties

- The Ultimate Goal is to measure material properties that can be reliable by either test and correlate with one another
- The intermediate results are used in the design of structures which are:
  - In a much larger structure,
  - Increasing the randomness of crack initiation,
  - Results are affected by reinforcement, and loading geometry

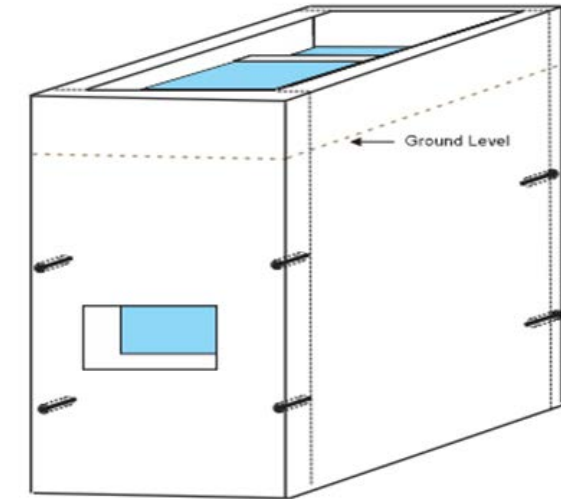


## SRP Testing (Hydraulic structures, PP fibers)

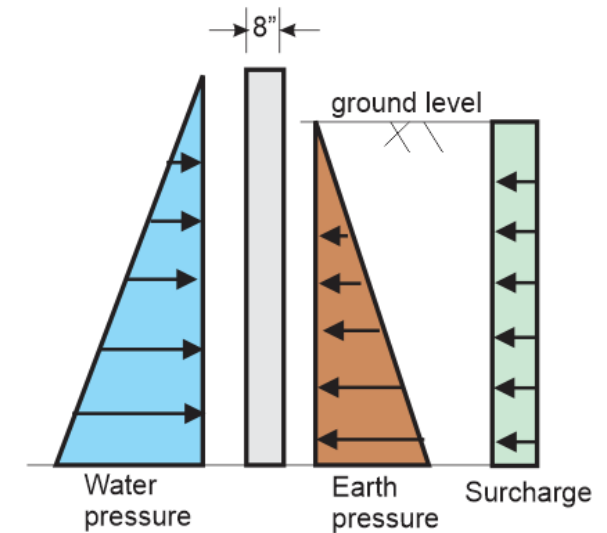
### Conventional Precast Section Used



- 75% Concrete ignored in design
- Need more efficient use of concrete
- Sustainability and Carbon footprint?
- Need corrosion control, durability and higher service life



Precast Panel water tank



Typical Loading

## Experimental Program

Panels Studied: FRC, HRC and Conventional

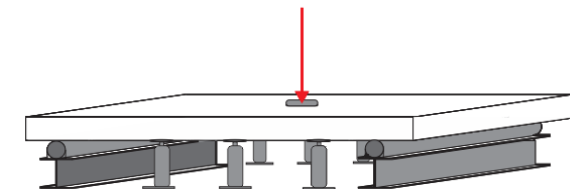
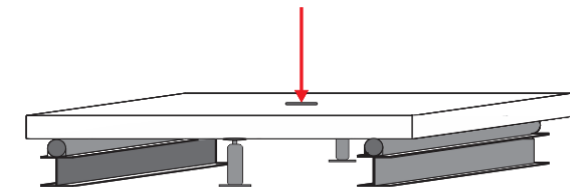
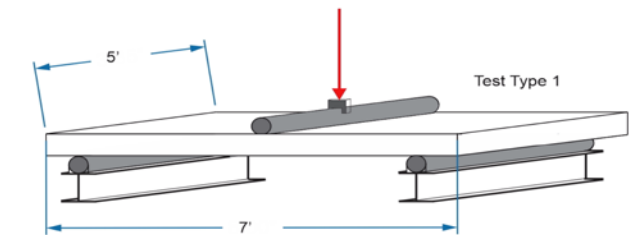
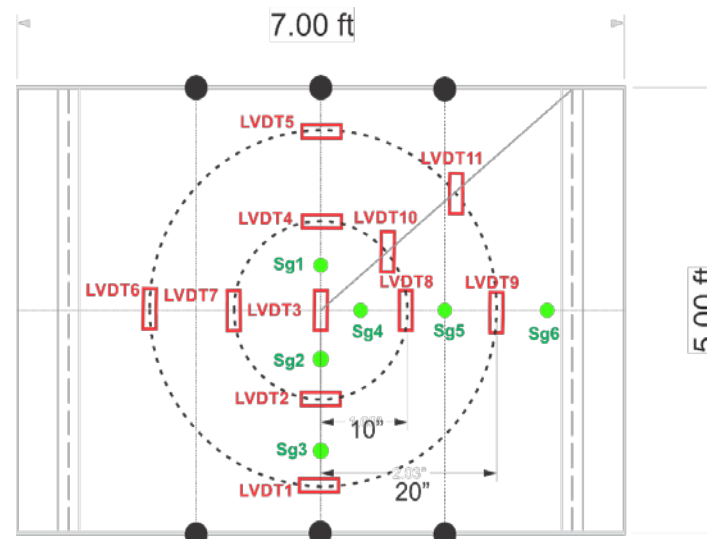
Tested for:

- Multiple point supports  
(Higher redundancy with multiple load paths)
- Also tested for 3PB
- Point Load Fatigue



Instrumentation:

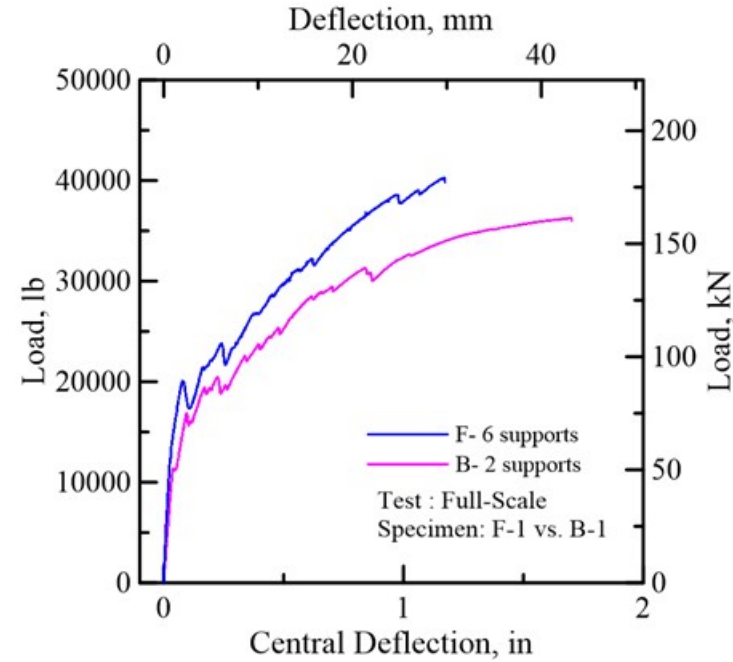
- 13 LVDT  
(complete 2D profile)
- 8 Strain Gages
- DIC
- String Sensors



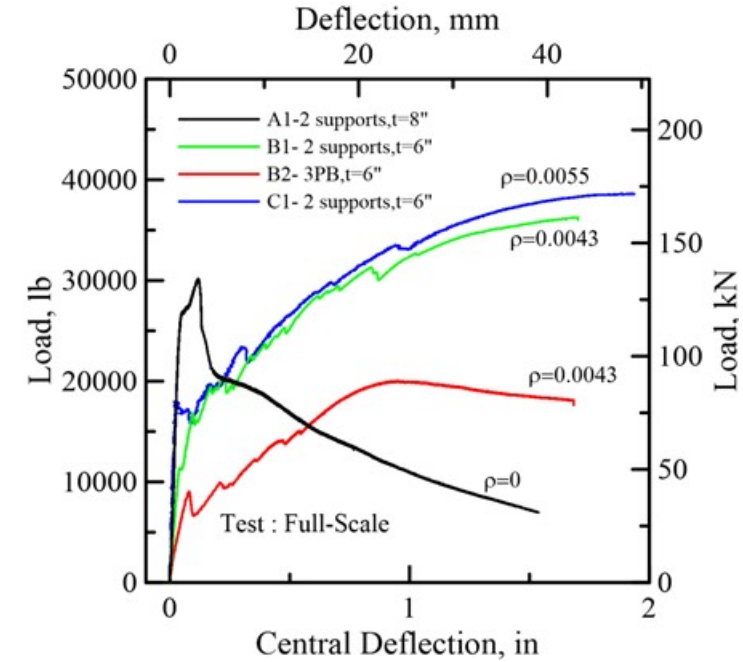
## Results



Multiple Cracks in HRC

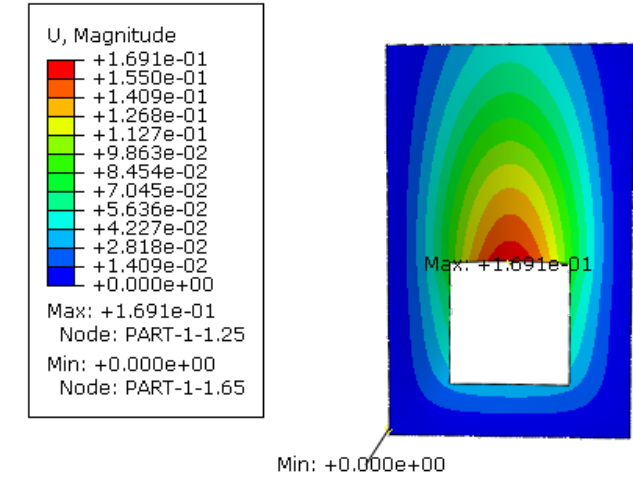
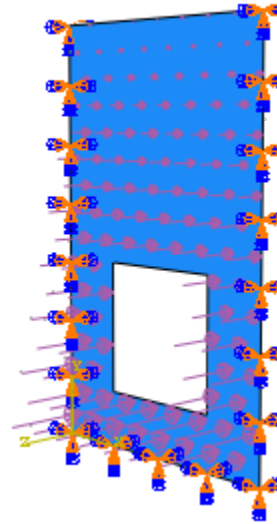
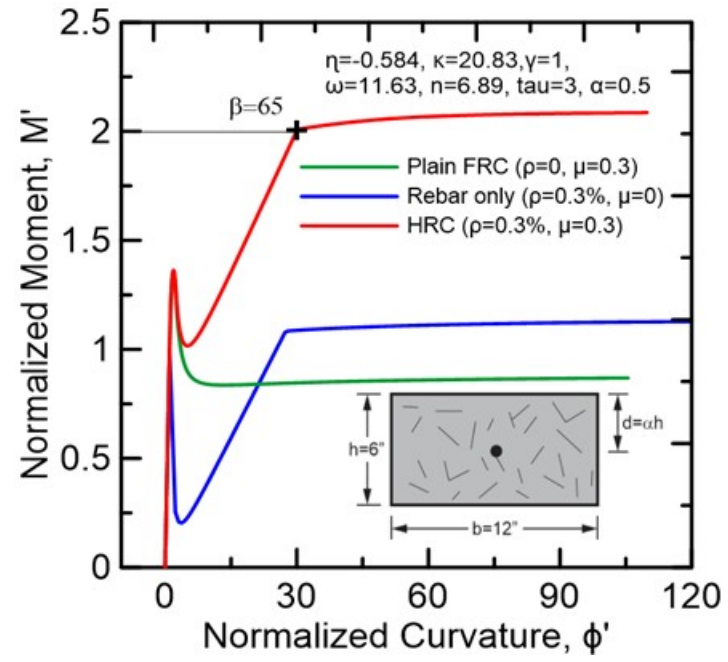


Effects of support



Difference in FRC, HRC, reinforcement ratio and 3PB

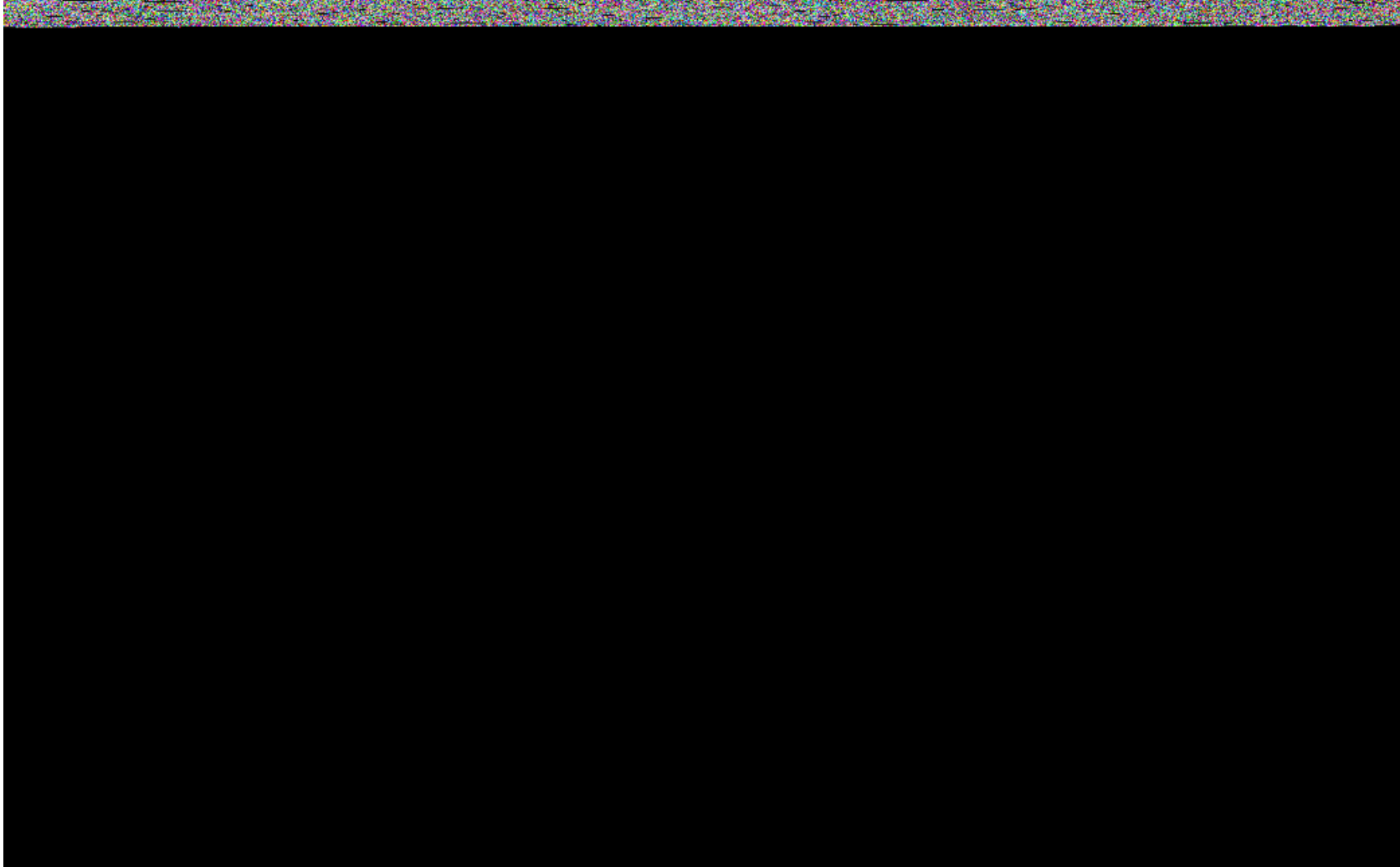
## Miscellaneous Slides



Maximum Displacement, mesh size = 1 inch,  
#node=26266, Q8 element

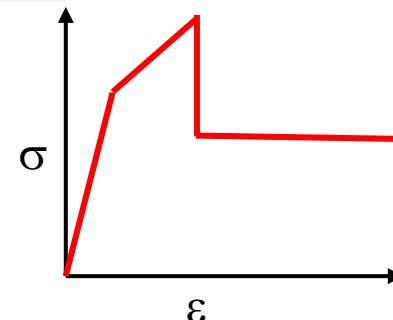
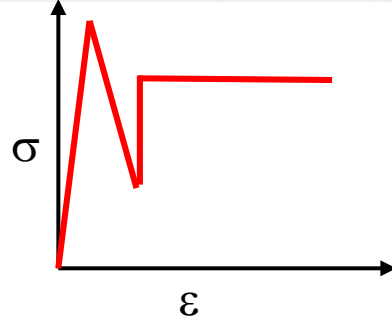
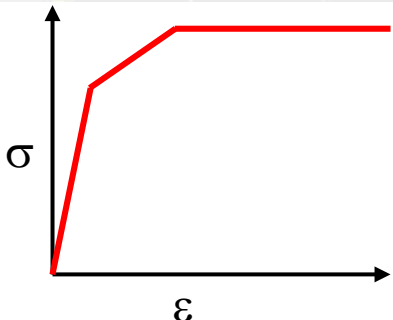
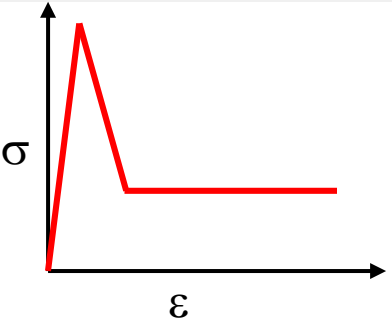
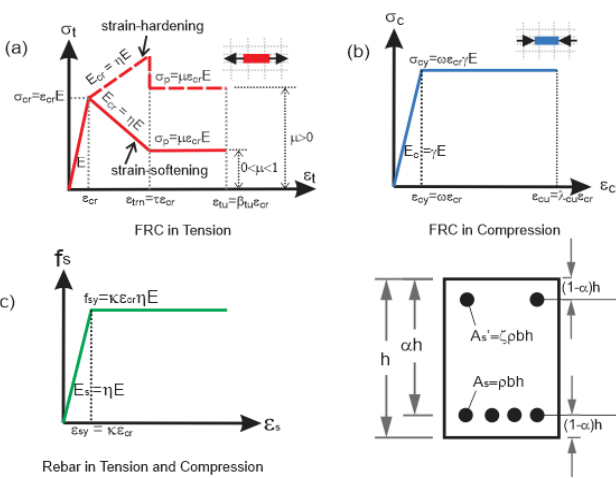
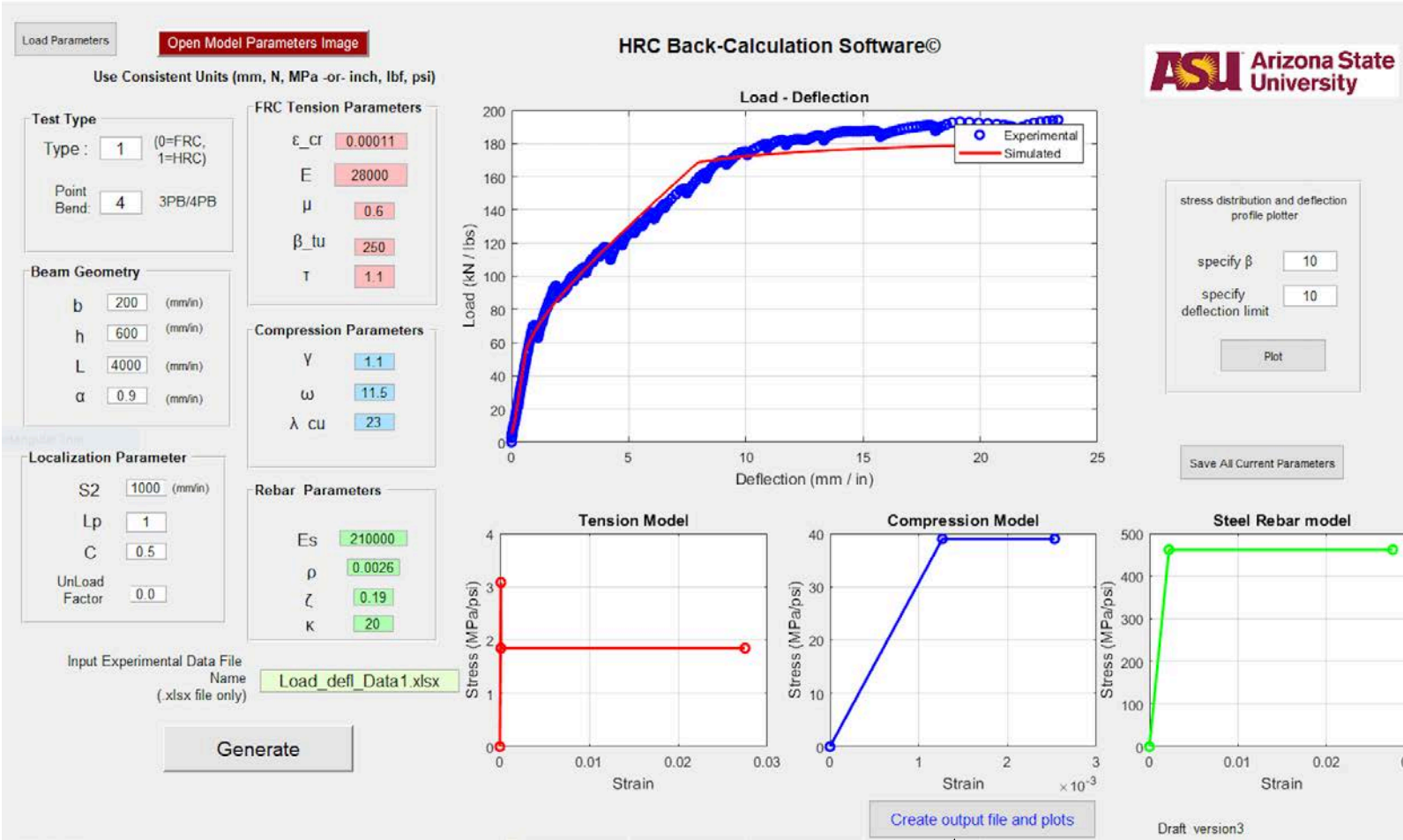
## Software Based HRC Design

### HRC Model



# HRC Back-Calculation software (Matlab)

Output will produce 12 plots with an output file



# Supporting software and guides

- An excel spreadsheet with associated uses manual that computes the material properties for three- and four-point bending samples
- Versions for Hybrid Design, (with tension reinforcement ) compression P-M diagram are also available for download
- ACI 544 Committee web page [https://www.concrete.org/committees/directoryofcommittees/acommitteehome.aspx?committee\\_code=0000544-00](https://www.concrete.org/committees/directoryofcommittees/acommitteehome.aspx?committee_code=0000544-00)
- ResearchGate Data Barzin Mobasher [https://www.researchgate.net/profile/Barzin\\_Mobasher](https://www.researchgate.net/profile/Barzin_Mobasher)
- A Spreadsheet-Based Inverse Analysis Procedure for Flexural Specimens – Strain Softening/Hardening Behavior
- DOI 10.13140/RG.2.1.4409.2882
- [https://www.researchgate.net/publication/283722776\\_A\\_Spreadsheet-Based\\_Inverse\\_Analysis\\_Procedure\\_for\\_Flexural\\_Specimens - Strain SofteningHardening Behavior](https://www.researchgate.net/publication/283722776_A_Spreadsheet-Based_Inverse_Analysis_Procedure_for_Flexural_Specimens_-_Strain_SofteningHardening_Behavior)